

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #14 – Corporate Mental Wellness & Counseling Platform

Complete Solution

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall generate an anonymized pseudo-ID for each employee account so counseling can be booked without revealing corporate email or employee ID to the therapist.	High	Therapist-facing booking records show the pseudo-ID only; corporate email and employee ID are absent.	Protects identity and supports confidential counseling.
FR-002	Functional	The system shall allow an employee to complete a stress-level assessment and view the resulting personal wellness indicator.	High	A completed assessment produces a wellness indicator visible only to the employee and stored against the pseudo-ID.	Enables personal wellness assessment.
FR-003	Functional	The system shall allow an employee to view licensed therapists and book an available confidential counseling session.	High	An available slot can be selected and booked, and confirmation shows the appointment date and time.	Provides the core counseling-booking function.
FR-004	Functional	The system shall allow a licensed therapist to view scheduled sessions using the employee pseudo-ID and update session status.	High	The therapist can view the pseudo-ID and mark a session scheduled, completed, or cancelled.	Lets therapists manage appointments without employee identity.
FR-005	Functional	The system shall allow an employee to track personal wellness indicators and upcoming counseling sessions through a private dashboard.	Medium	The dashboard shows the employee's wellness history and upcoming sessions, with no other employee data visible.	Supports ongoing wellness tracking.
NFR-001	Nonfunctional – Performance & Security	The appointment scheduler shall handle timezone conversions accurately across global office locations and remain responsive under peak load.	High	Timezone test cases preserve the intended appointment instant, and benchmark tests meet the defined response target under simulated peak load.	Prevents scheduling errors across locations.
NFR-002	Nonfunctional – Privacy & Security	The system shall protect employee identity and confidential wellness information from unauthorized access.	High	Security tests confirm therapists see only the pseudo-ID and authorized session data, while unauthorized users cannot access another employee's wellness data.	Confidentiality is central to the platform.

2. Actors and Primary Use Cases

Actors

- Corporate Employee
- Licensed Therapist

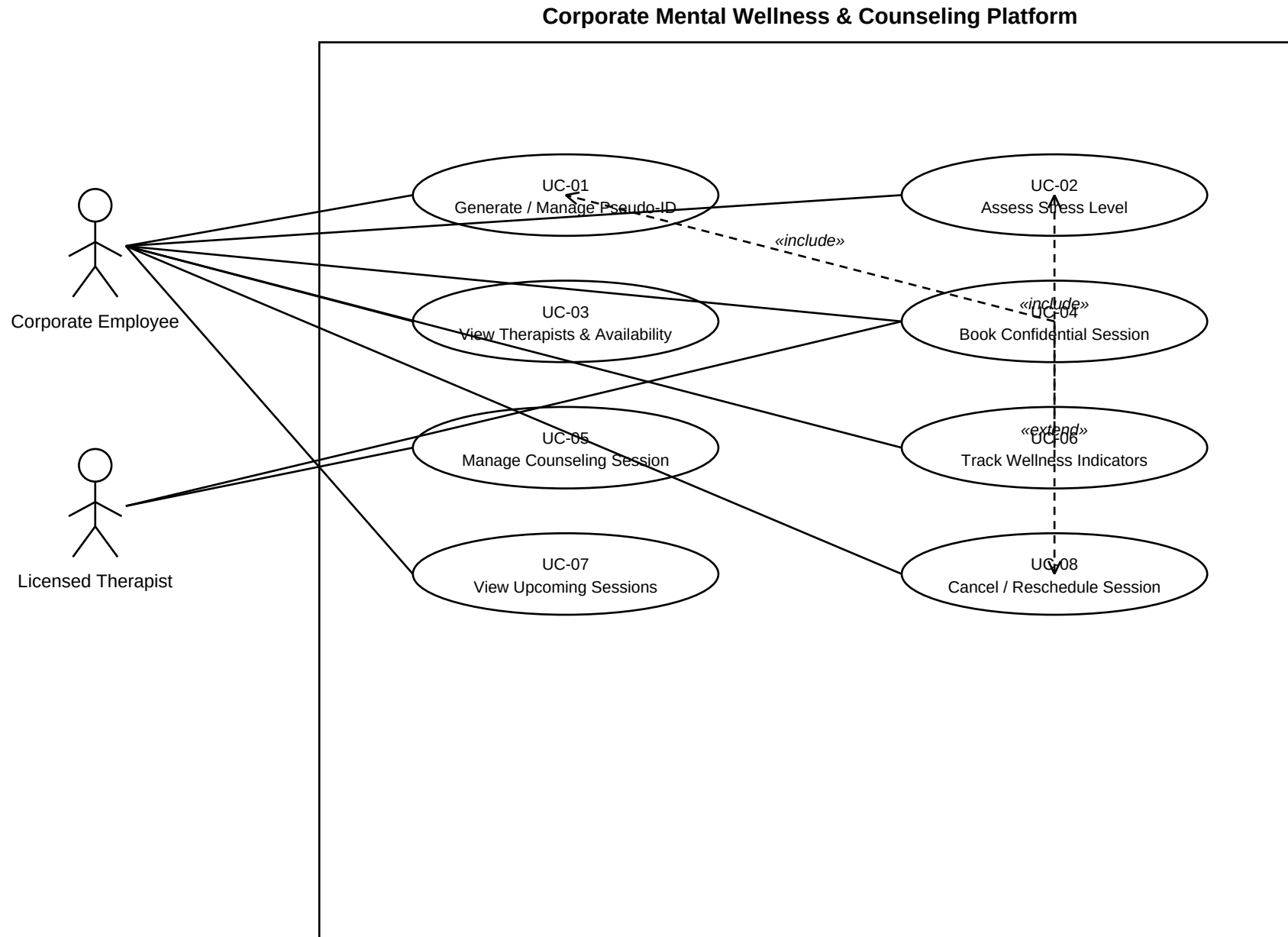
Use Cases

ID	Use Case	Purpose
UC-01	Generate / Manage Pseudo-ID	Anonymized identity for confidential interactions.
UC-02	Assess Stress Level	Complete assessment and receive wellness indicator.
UC-03	View Therapists & Availability	View therapists and available counseling slots.
UC-04	Book Confidential Session	Book a counseling appointment.
UC-05	Manage Counseling Session	Therapist views sessions and updates status.
UC-06	Track Wellness Indicators	Privately view wellness indicators.
UC-07	View Upcoming Sessions	View confirmed appointments.
UC-08	Cancel / Reschedule Session	Optionally change or cancel an appointment.

3. UML Use-Case Diagram

The diagram includes both required relationship types: «include» and «extend».

UML Use-Case Diagram – Corporate Mental Wellness & Counseling Platform



4. Use-Case Flow Specification – Book Confidential Counseling Session

Preconditions

- Employee is authenticated.
- Licensed therapists and available slots exist.
- Employee pseudo-ID is available.

Postconditions

- Confirmed appointment is stored against the pseudo-ID.
- Employee receives confirmation.
- Therapist sees the session without corporate email or employee ID.

Main Success Scenario

1. Employee signs in to the counseling platform.
2. System retrieves the employee's anonymized pseudo-ID; corporate email and employee ID are not exposed to the therapist.
3. Employee views available licensed therapists and counseling time slots.
4. Employee selects a therapist and an available slot.
5. System validates the slot and performs the required timezone conversion.
6. System creates the confidential appointment using the pseudo-ID.
7. System displays the appointment confirmation to the employee.
8. Therapist dashboard shows the scheduled session using the pseudo-ID only.
9. The use case ends successfully.

Alternate Flow

- A1. Selected slot unavailable: the system informs the employee and displays alternative available slots.
- A2. Timezone conversion issue: the system recalculates and displays the appointment in the selected local timezone before confirmation.
- A3. Employee cancels before confirmation: the system stops booking and returns to the available-session view.